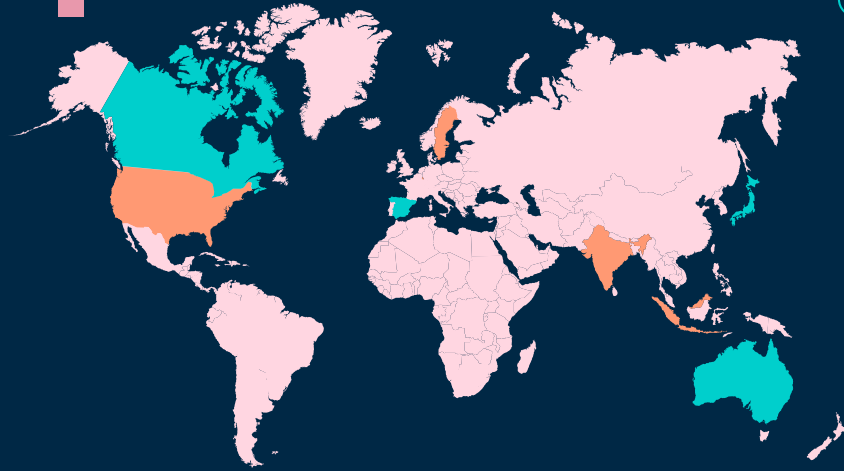




# Country Life Expectancy

# Understanding Life Expectancy

- **Problem statement**

Life expectancy is a vital measure of a nation's healthcare effectiveness. However, it varies widely due to complex **economic** and **environmental factors** **Countries. So, understanding the key contributors to live expectancy can help policymakers develop focused strategies to enhance public health outcomes.**

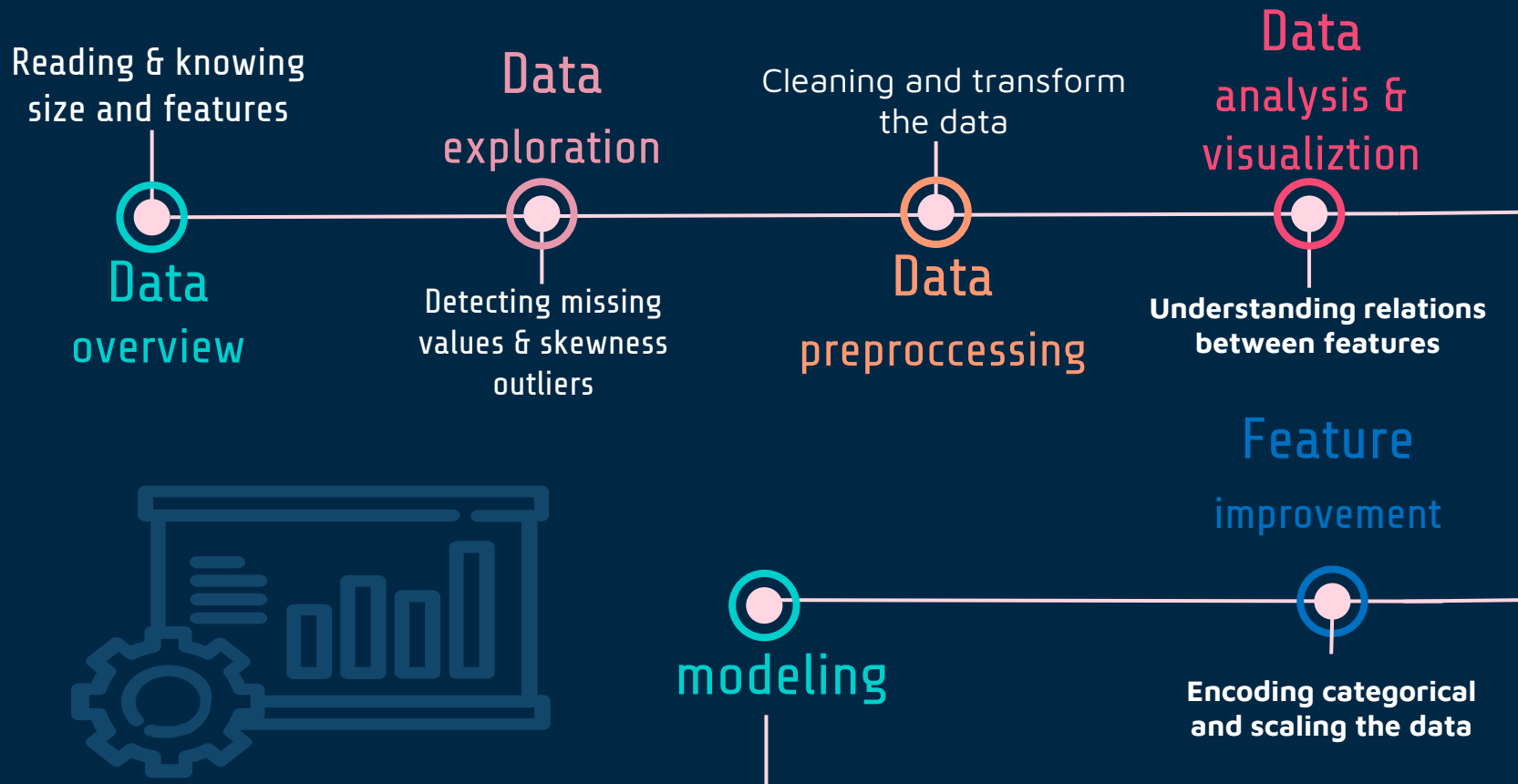
- **Objective**

Predict **Life Expectancy (Years)** for nations using machine learning to uncover complex relationships between health, economic, demographic, and behavioral factors.

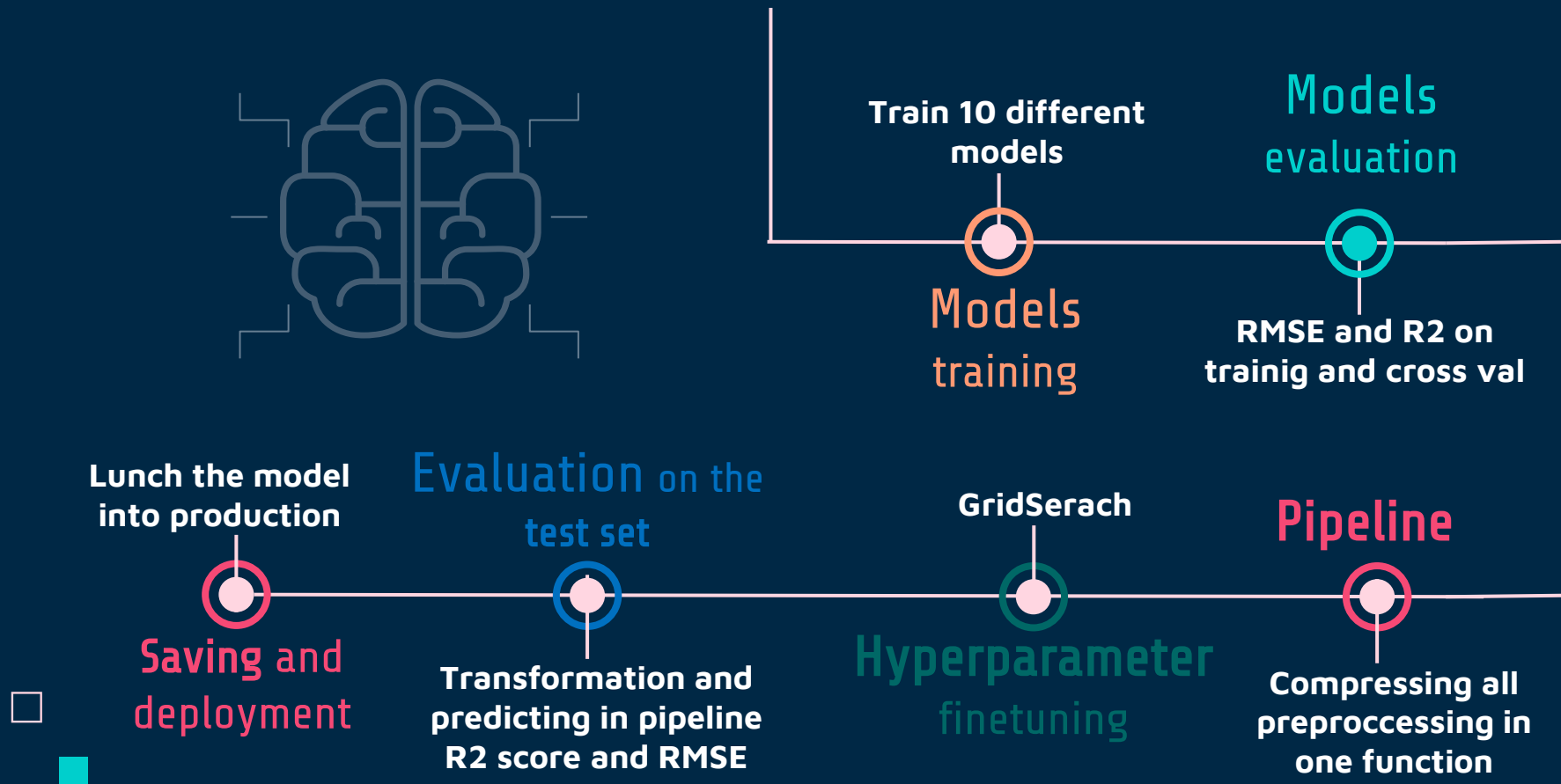
- **Key Features**

**Health indicators (e.g., vaccination coverage, mortality rates, HIV/AIDS prevalence),  
Economic metrics (e.g., GDP, health expenditure),  
Demographic data (e.g., total population, thinness percentage),  
Behavioral factors (e.g., alcohol consumption rate).**

# OUR PROCESS



# OUR PROCESS



# Data overview

Reading & knowing size  
and features

# Schema To understand the data

|                                         |                                                                                           |
|-----------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Unnamed: 0</b>                       | An index or unique identifier for the rows in the dataset, often auto-generated.          |
| <b>Nation</b>                           | The name of the nation or country corresponding to the data entry.                        |
| <b>Survey_Year</b>                      | The year when the survey or data collection occurred.                                     |
| <b>Country_Category</b>                 | The economic or regional classification of the country (e.g., 'Developing', 'Developed'). |
| <b>Mortality_Adults</b>                 | The adult mortality rate per 1000 adults aged 15-60.                                      |
| <b>Infant_Deaths_Count</b>              | The total number of infant (children under 1 year) deaths per year.                       |
| <b>Alcohol_Consumption_Rate</b>         | The per capita alcohol consumption rate in liters per year.                               |
| <b>Expenditure_Percentage_GDP</b>       | The percentage of the Gross Domestic Product (GDP) spent on health.                       |
| <b>Hepatitis_B_Vaccination_Coverage</b> | The percentage of the population vaccinated against Hepatitis B (B التهاب الكبد).         |
| <b>Measles_Infection_Count</b>          | The total number of reported measles cases (الحصبة).                                      |
| <b>Body_Mass_Index_Avg</b>              | The average body mass index (BMI) of the population.                                      |
| <b>Polio_Vaccination_Coverage</b>       | The percentage of the population vaccinated against Polio (تطعيم شلل الاطفال).            |
| <b>Total_Health_Expenditure</b>         | The total health expenditure per capita (in USD).                                         |
| <b>Diphtheria_Vaccination_Coverage</b>  | The percentage of the population vaccinated against Diphtheria (الديفتيريا).              |
| <b>HIV_AIDS_Prevalence_Rate</b>         | The prevalence rate of HIV/AIDS in the population as a percentage (الايذز).               |
| <b>Gross_Domestic_Product</b>           | The Gross Domestic Product (GDP) per capita (in USD).                                     |
| <b>Total_Population</b>                 | The total population of the country.                                                      |
| <b>Thinness</b>                         | The percentage of the population classified as thin (low BMI).                            |
| <b>Life_Expectancy_Years</b>            | The average number of years a person is expected to live.                                 |

# 01

## Data Overview

Using info to look  
through the data

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 2938 entries, 0 to 2937
```

```
Data columns (total 19 columns):
```

| #  | Column                           | Non-Null Count | Dtype   |
|----|----------------------------------|----------------|---------|
| 0  | Unnamed: 0                       | 2938 non-null  | int64   |
| 1  | Nation                           | 2937 non-null  | object  |
| 2  | Survey_Year                      | 2936 non-null  | float64 |
| 3  | Country_Category                 | 2935 non-null  | object  |
| 4  | Mortality_Adults                 | 2925 non-null  | float64 |
| 5  | Infant_Deaths_Count              | 2938 non-null  | int64   |
| 6  | Alcohol_Consumption_Rate         | 2744 non-null  | float64 |
| 7  | Expenditure_Percentage_GDP       | 2938 non-null  | float64 |
| 8  | Hepatitis_B_Vaccination_Coverage | 2385 non-null  | float64 |
| 9  | Measles_Infection_Count          | 2936 non-null  | float64 |
| 10 | Body_Mass_Index_Avg              | 2904 non-null  | float64 |
| 11 | Polio_Vaccination_Coverage       | 2919 non-null  | float64 |
| 12 | Total_Health_Expenditure         | 2711 non-null  | float64 |
| 13 | Diphtheria_Vaccination_Coverage  | 2919 non-null  | float64 |
| 14 | HIV_AIDS_Prevalence_Rate         | 2938 non-null  | float64 |
| 15 | Gross_Domestic_Product           | 2490 non-null  | float64 |
| 16 | Total_Population                 | 2286 non-null  | float64 |
| 17 | Thinness                         | 2904 non-null  | float64 |
| 18 | Life_Expectancy_Years            | 2928 non-null  | float64 |

```
dtypes: float64(15), int64(2), object(2)
```

```
memory usage: 436.2+ KB
```

01

# Data Overview

Describe to get statistics  
about the data

|       | Unnamed: 0  | Survey_Year | Mortality_Adults | Infant_Deaths_Count | Alcohol_Consumption_Rate | Expenditure_Percentage_GDP | Hepatitis_B_Vaccination_Coverage | Mea |
|-------|-------------|-------------|------------------|---------------------|--------------------------|----------------------------|----------------------------------|-----|
| count | 2938.000000 | 2936.000000 | 2925.000000      | 2938.000000         | 2744.000000              | 2938.000000                | 2385.000000                      |     |
| mean  | 1468.500000 | 2007.52282  | 164.865299       | 30.303948           | 4.602861                 | 738.251295                 | 80.940461                        |     |
| std   | 848.271871  | 4.61257     | 124.316868       | 117.926501          | 4.052413                 | 1987.914858                | 25.070016                        |     |
| min   | 0.000000    | 2000.00000  | 1.000000         | 0.000000            | 0.010000                 | 0.000000                   | 1.000000                         |     |
| 25%   | 734.250000  | 2004.00000  | 74.000000        | 0.000000            | 0.877500                 | 4.685343                   | 77.000000                        |     |
| 50%   | 1468.500000 | 2008.00000  | 144.000000       | 3.000000            | 3.755000                 | 64.912906                  | 92.000000                        |     |
| 75%   | 2202.750000 | 2012.00000  | 228.000000       | 22.000000           | 7.702500                 | 441.534144                 | 97.000000                        |     |
| max   | 2937.000000 | 2015.00000  | 723.000000       | 1800.000000         | 17.870000                | 19479.911610               | 99.000000                        |     |



01

# Data Overview

**Shape (2938,19 )**

**Some of the data are Categorical  
Nation and Country Category  
and other are numerical  
so make two lists to define them**

**Finally we spitted the data into train and  
test set**



# Data exploration

Detecting missing  
values & skewness  
outliers

02

# Data Exploration

**Inspect the data for :**

- **missing values**
- **Outliers**
- **Duplicates**
- **Skewness**



# 02

## Data Exploration

### Missing values

There are missing values  
With different ranges  
In the dataset

```
train_set.isna().sum()
```

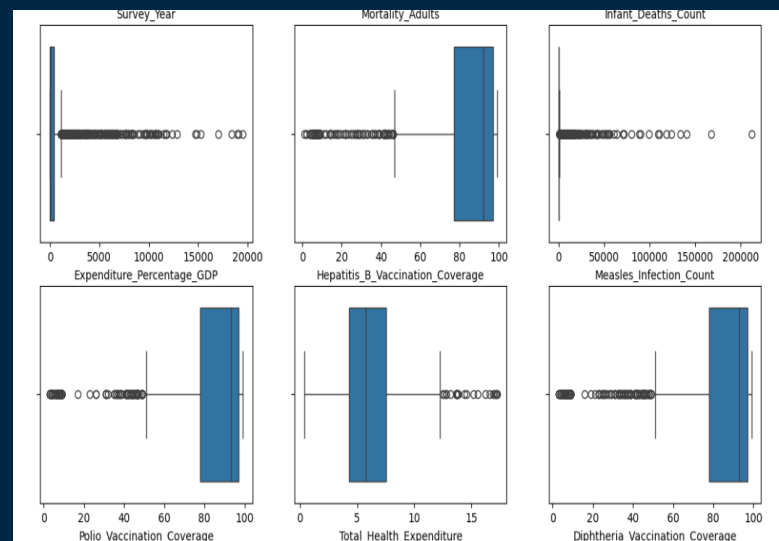
|                                  |     |
|----------------------------------|-----|
| Nation                           | 1   |
| Survey_Year                      | 0   |
| Country_Category                 | 3   |
| Mortality_Adults                 | 9   |
| Infant_Deaths_Count              | 0   |
| Alcohol_Consumption_Rate         | 150 |
| Expenditure_Percentage_GDP       | 0   |
| Hepatitis_B_Vaccination_Coverage | 449 |
| Measles_Infection_Count          | 2   |
| Body_Mass_Index_Avg              | 26  |
| Polio_Vaccination_Coverage       | 14  |
| Total_Health_Expenditure         | 178 |
| Diphtheria_Vaccination_Coverage  | 14  |
| HIV_AIDS_Prevalence_Rate         | 0   |
| Gross_Domestic_Product           | 354 |
| Total_Population                 | 521 |
| Thinness                         | 26  |
| Life_Expectancy_Years            | 7   |
| dtype: int64                     |     |

02

# Data Exploration

## Outliers

Using Boxplot there are a lot of outliers appear in each column

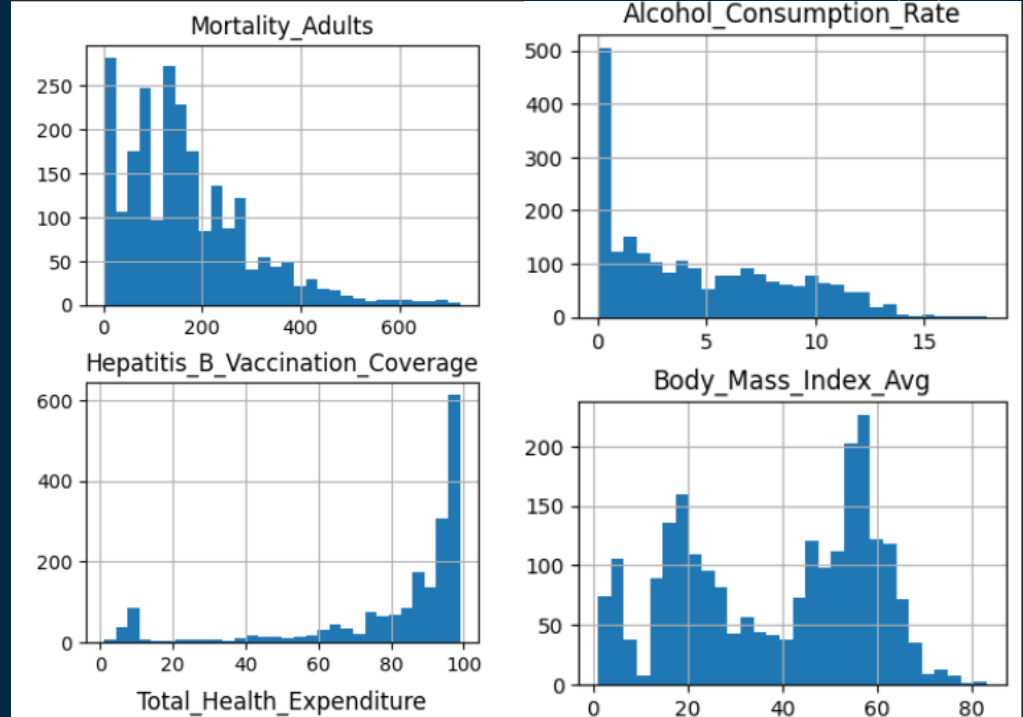


02

# Data Exploration

## Skewness

Using Histogram,  
there skewness in the  
data some to the left  
other to the right

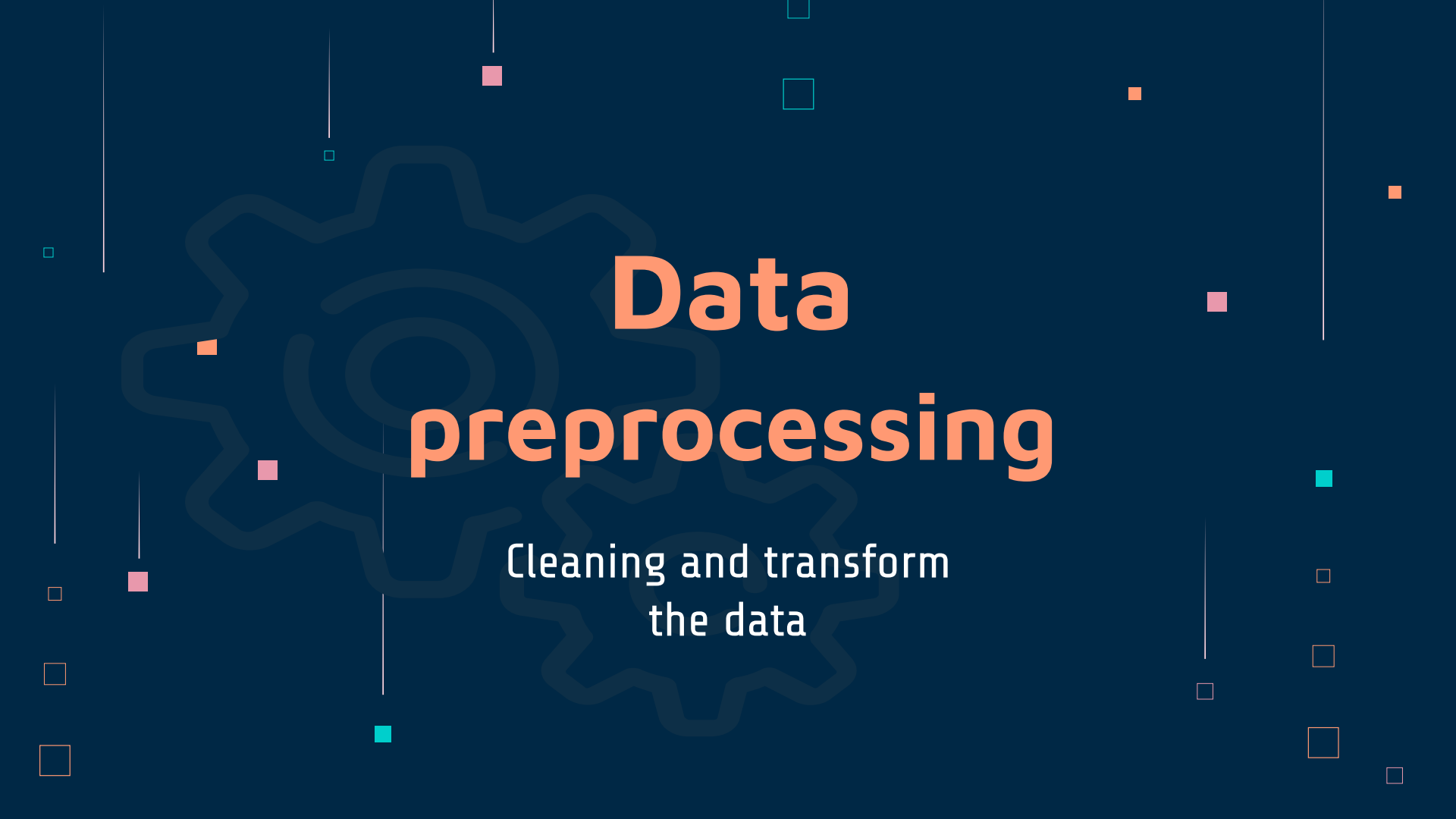


# 02

## Data

### Exploration conclusion

- **Unnamed Column:** It is useless, so we dropped it.
- **Data Types:** Defined the train\_set into categorical and numerical types.
- **Duplicates:** There are no duplicate entries in the train\_set.
- **Missing Values:** Most columns have a small number of missing values, but:
  - **Gross Domestic Product:** 15.25% missing
  - **Total Population:** 22.19% missing
  - **Hepatitis B Vaccination Coverage:** 18.82% missing
  - **Total Health Expenditure:** 7.73% missing
  - **Alcohol Consumption Rate:** 6.60% missing
- **Skewness:** Many columns exhibit skewness, which should be handled appropriately.
- **Outliers:** Detected many outliers through boxplots. These should be removed or handled.
- **Scaling:** The train\_set has varying ranges, so it needs to be scaled.



# Data preprocessing

Cleaning and transform  
the data



# 03

## Data preprocessing

preprocessing data through :

- Handling wrong values
- Replacing missing values
- Handling outliers
- Handling skewness

# 03

## Data preprocessing

### Handling wrong values

- Some percentage column have wrong values that are beyond 100 so we replaced it by Nan

```
Nation 1
Survey_Year 0
Country_Category 3
Mortality_Adults 9
Infant_Deaths_Count 0
Alcohol_Consumption_Rate 150
Expenditure_Percentage_GDP 0
Hepatitis_B_Vaccination_Coverage 449
Measles_Infection_Count 2
Body_Mass_Index_Avg 26
Polio_Vaccination_Coverage 14
Total_Health_Expenditure 178
Diphtheria_Vaccination_Coverage 14
HIV_AIDS_Prevalence_Rate 0
Gross_Domestic_Product 354
Total_Population 521
Thinness 26
Life_Expectancy_Years 7
dtype: int64
```

# 03

## Data preprocessing

### Handling wrong values

The Expenditure\_percentage\_GDP has  
Almost half of the values Non values  
So we dropped it

|                                  |           |
|----------------------------------|-----------|
| Nation                           | 0.042553  |
| Survey_Year                      | 0.000000  |
| Country_Category                 | 0.127660  |
| Mortality_Adults                 | 0.382979  |
| Infant_Deaths_Count              | 0.000000  |
| Alcohol_Consumption_Rate         | 6.382979  |
| Expenditure_Percentage_GDP       | 44.680851 |
| Hepatitis_B_Vaccination_Coverage | 19.106383 |
| Measles_Infection_Count          | 0.085106  |
| Body_Mass_Index_Avg              | 1.106383  |
| Polio_Vaccination_Coverage       | 0.595745  |
| Total_Health_Expenditure         | 7.574468  |
| Diphtheria_Vaccination_Coverage  | 0.595745  |
| HIV_AIDS_Prevalence_Rate         | 0.000000  |
| Gross_Domestic_Product           | 15.063830 |
| Total_Population                 | 22.170213 |
| Thinness                         | 1.106383  |
| Life_Expectancy_Years            | 0.297872  |

dtype: float64

# 03

## Data preprocessing

### Replacing missing values

- Filled the missing values using pandas fillna with median
- We dropped the row with missing value in nation
- In country category there were some countries that hasn't been provided searched in the internet and filled it

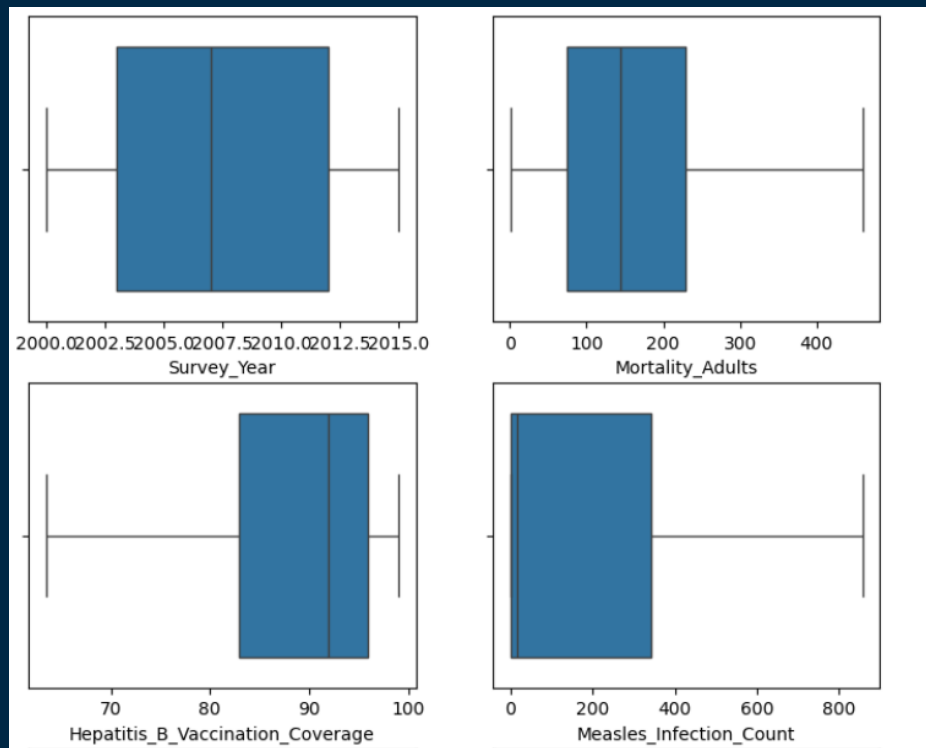
```
Nation 0
Survey_Year 0
Country_Category 0
Mortality_Adults 0
Infant_Deaths_Count 0
Alcohol_Consumption_Rate 0
Hepatitis_B_Vaccination_Coverage 0
Measles_Infection_Count 0
Body_Mass_Index_Avg 0
Polio_Vaccination_Coverage 0
Total_Health_Expenditure 0
Diphtheria_Vaccination_Coverage 0
HIV_AIDS_Prevalence_Rate 0
Gross_Domestic_Product 0
Total_Population 0
Thinness 0
Life_Expectancy_Years 0
dtype: int64
```

## 03

# Data preprocessing

## Handling outliers

- We used `clip()` func to remove data that is over first and third quartile
- Then we use `sns.boxplot()` for visualization



03

# Data preprocessing

## Handling skewness

**When we handle skewing, we must take care of positive skewness and negative skewness**

## 03

# Data preprocessing

## Positive skewness

We have tested 4 transformers to decide which will have best result to solve positive skewness which is :

Yeo-Johnson

|             | Mortality_Adults | Infant_Deaths_Count | Alcohol_Consumption_Rate | Measles_Infection_Count | HIV_AIDS_Prevalence_Rate | Gross_Domestic_Product | Total_Population | Thinness |
|-------------|------------------|---------------------|--------------------------|-------------------------|--------------------------|------------------------|------------------|----------|
| Log         | -1.221521        | 0.323474            | -0.327322                | 0.179176                | 1.094774                 | -0.704610              | -1.303018        | 0.029753 |
| Square Root | -0.073241        | 0.686234            | -0.150476                | 0.838711                | 1.044908                 | 0.608273               | 0.651534         | 0.382103 |
| Yeo-Johnson | -0.116153        | 0.128084            | -0.086623                | 0.088557                | 0.775877                 | -0.070137              | -0.121003        | 0.005102 |
| Quantile    | 1.294335         | 0.000955            | -1.660091                | 0.152245                | 0.791542                 | 1.423249               | 1.401660         | 0.881403 |
| Original    | 0.773016         | 1.264455            | 0.629462                 | 1.185867                | 1.204355                 | 1.167220               | 1.200303         | 1.096385 |

03

# Data preprocessing

## Negative skewness

We have tested 6 transformers to decide which will have best result to solve negative skewness which is :

Square

|             | Hepatitis_B_Vaccination_Coverage | Polio_Vaccination_Coverage | Diphtheria_Vaccination_Coverage |
|-------------|----------------------------------|----------------------------|---------------------------------|
| Exponential | 2.551724                         | 1.890686                   | 1.960884                        |
| Square      | -0.995997                        | -1.039938                  | -1.056863                       |
| Cube        | -0.861488                        | -0.846765                  | -0.858272                       |
| Reciprocal  | 1.366001                         | 1.710508                   | 1.724131                        |
| Log         | -1.248873                        | -1.480947                  | -1.500760                       |
| quantile    | -0.199443                        | 0.143486                   | 0.070557                        |
| Original    | -1.126357                        | -1.255547                  | -1.275907                       |

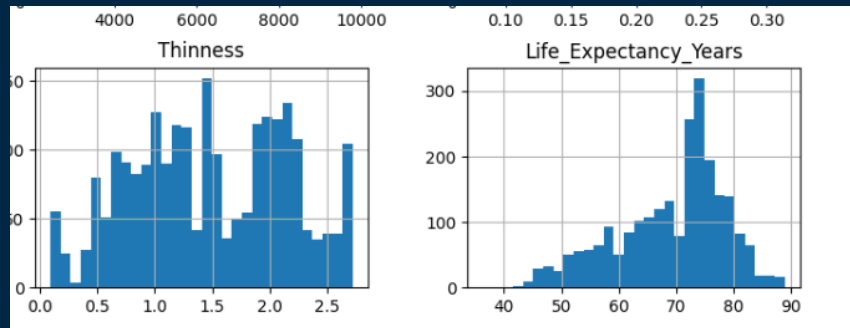


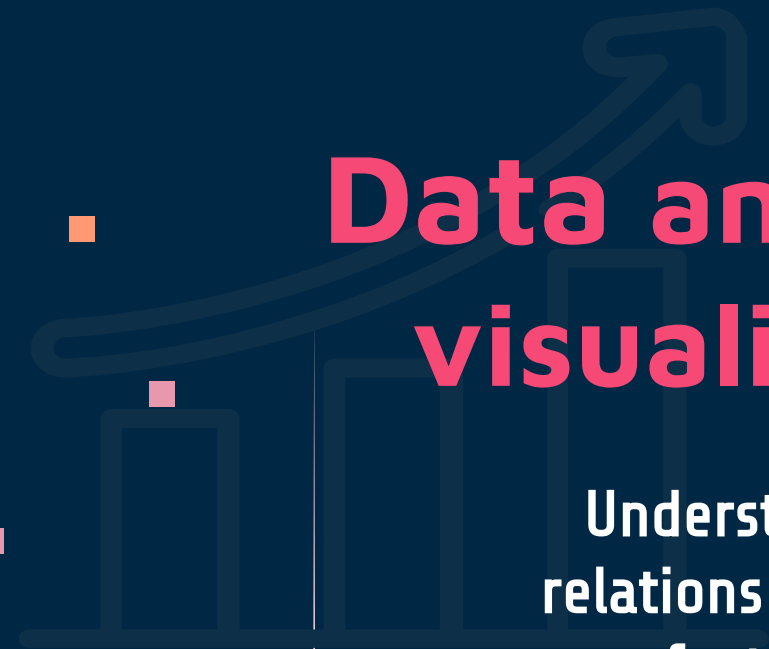
# 03

## Data preprocessing

### Handling skewness Result

- After positive and negative skewness



A faint, stylized background graphic featuring a bar chart with three bars of increasing height and a large, light-colored arrow pointing upwards and to the right, symbolizing growth and data analysis.

# Data analysis & visualization

Understanding  
relations between  
features

04

# Data visulaization

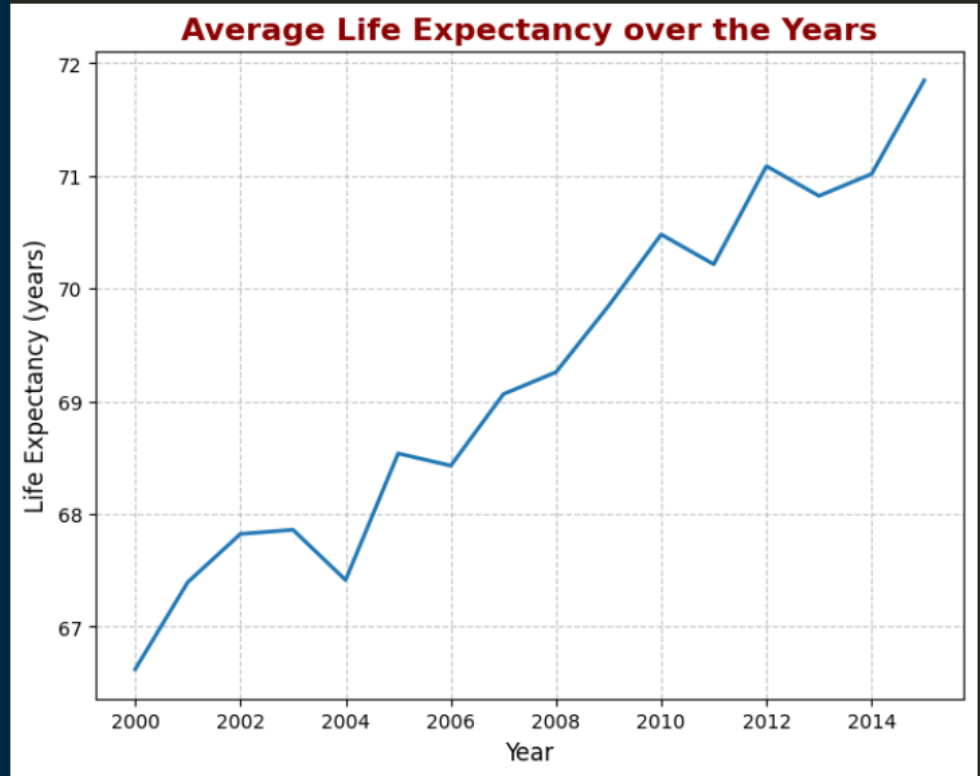
## Analysis and Visualization:

- **Data over years**
- **Sum Mortality Adults over the Years**
- **Developed vs Developing**
- **Correlation and multivariable analysis**

04

# Data visulaization

Data average over  
years



04

# Data visulaization

**Sum Mortality Adults  
over the Years**

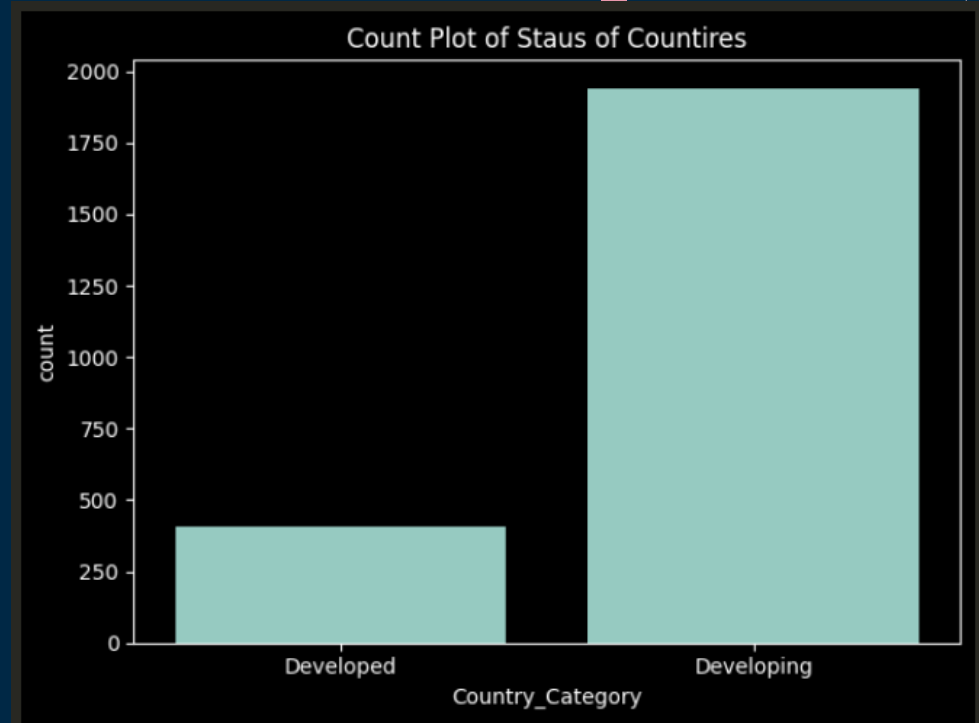


04

# Data visulaization

## Developed vs Developing

- Comparing by count of country\_category

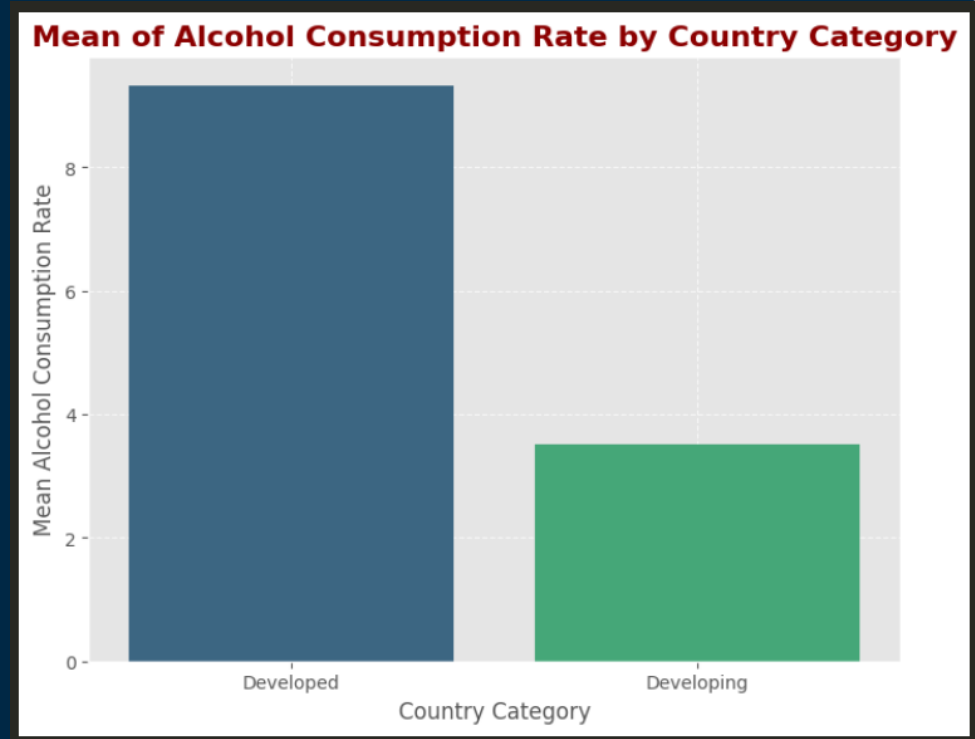


04

# Data visulaization

## Developed vs Developing

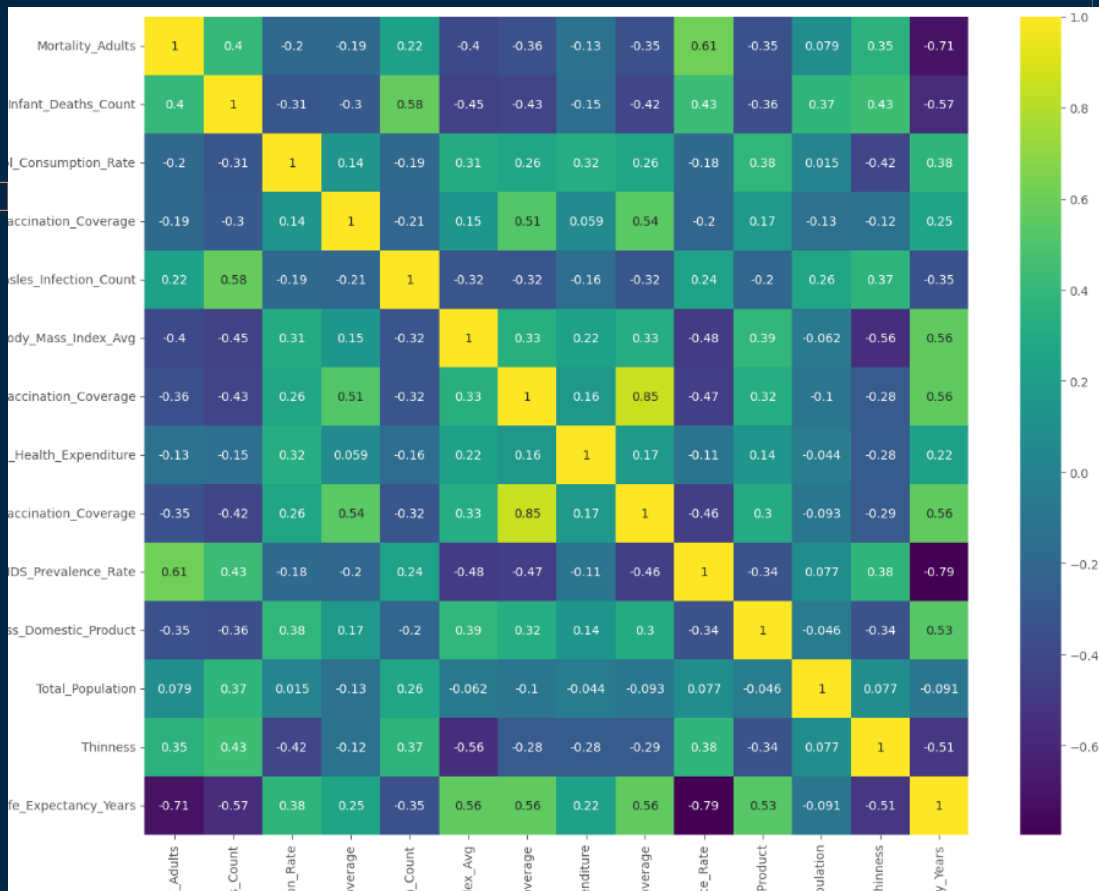
- Mean of Alcohol Consumption Rate by Country Category



## 04

Data  
visualizationCorrelation and  
multivariable analysis

- Correlation matrix to  
visualize correlation  
between columns and  
each other



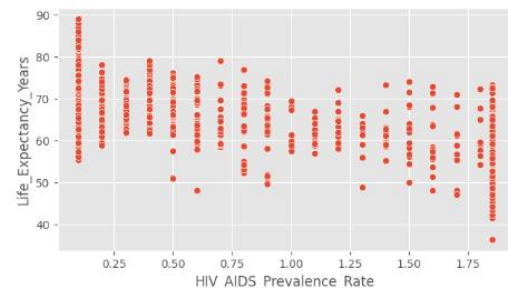
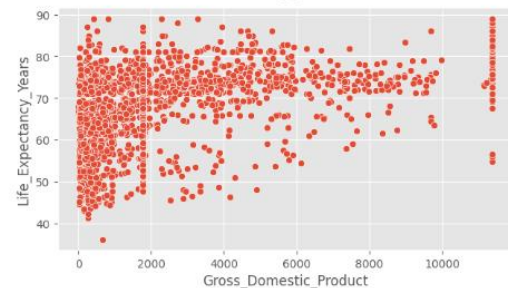
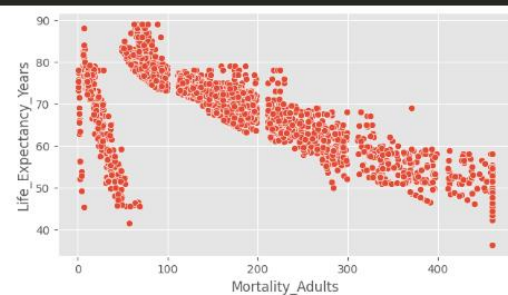
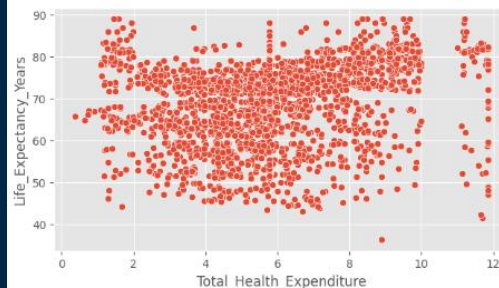
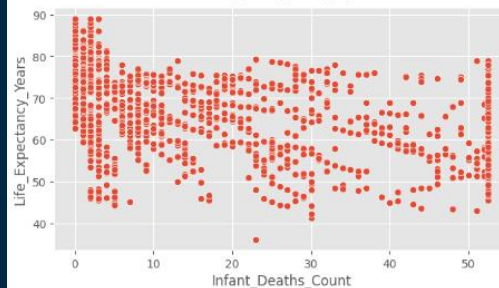
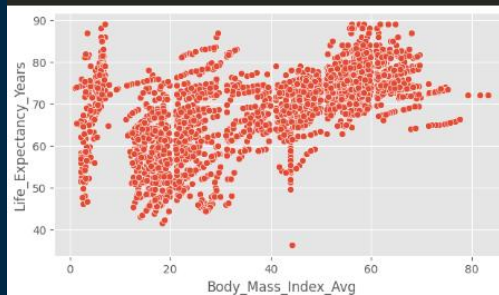


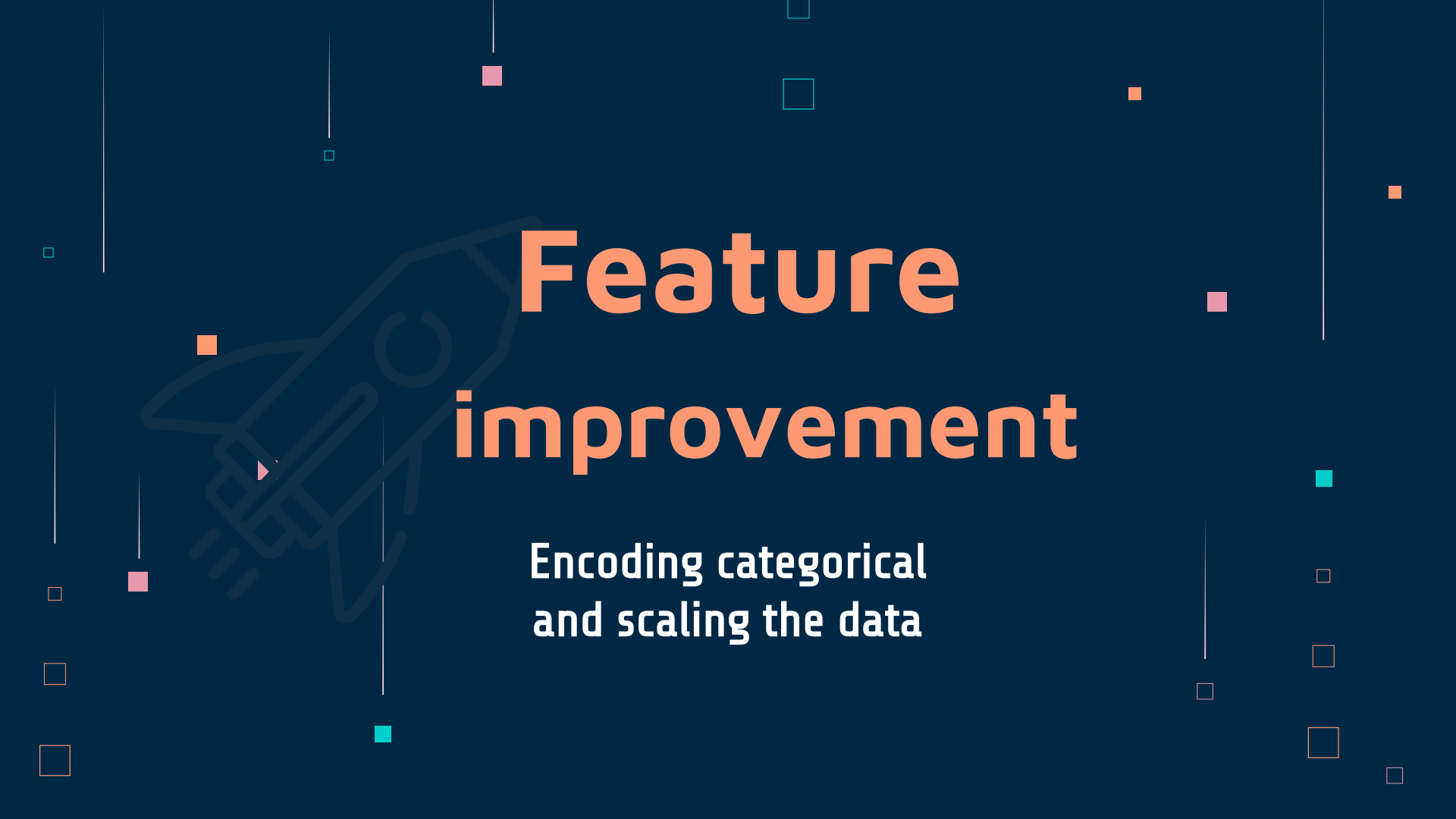
## 04

# Data visulaization

## Correlation and multivariable analysis

- Using scatterplot to show if there is any pattern in features with target column



The background is a dark blue gradient. It features a faint, light blue line-art illustration of a rocket ship on the left side, angled upwards. Scattered across the slide are numerous small squares in various colors (orange, pink, teal, light blue) and thin vertical white lines of varying lengths, creating a modern, tech-inspired aesthetic.

# Feature improvement

Encoding categorical  
and scaling the data

05

# Feature improvement

**Improving features:**

- **Converting categorial into numerical**
- **Scaling data**

# 05

## Feature improvement

### Converting categorical into numerical

- Using label encoder to replace categorical data with numerical

```
from sklearn.preprocessing import StandardScaler, LabelEncoder  
le = LabelEncoder()  
cat_cols = train_set.select_dtypes(include = 'object').columns  
for cols in cat_cols:  
    train_set[cols] = le.fit_transform(train_set[cols])
```

05

# Feature improvement

## Scaling data

- Using **StandardScaler()** to scale data

|      | Nation    | Survey_Year | Country_Category | Mortality_Adults | Infant_Deaths_Count | Alcohol_Consumption_Rate | Hepatitis_B_Vaccination_Coverage | Measles_Infection_Count | Body_Mass_Index_Avg | Polio_Vaccination_Coverage |
|------|-----------|-------------|------------------|------------------|---------------------|--------------------------|----------------------------------|-------------------------|---------------------|----------------------------|
| 456  | -0.371770 | 0.120589    | 0                | -0.882069        | -1.207603           | 1.036716                 | 0.366584                         | -1.138337               | 0.986362            | 0.857934                   |
| 462  | 0.357794  | -1.170885   | 1                | 0.723701         | 1.450872            | -1.227000                | 0.366584                         | 1.283716                | -1.206391           | -0.209686                  |
| 2172 | -1.666745 | 0.120589    | 1                | 1.523380         | 1.450872            | 0.799704                 | -1.605286                        | 0.969293                | -0.969473           | -1.395884                  |
| 2667 | -1.028377 | 1.196817    | 1                | -0.421840        | -0.603821           | -0.001250                | 0.564677                         | -1.138337               | 1.006525            | 0.615331                   |
| 381  | -0.244096 | 1.412063    | 0                | -0.896061        | -0.282802           | 0.884792                 | 0.665321                         | -1.138337               | 1.238402            | 0.615331                   |



# Models training

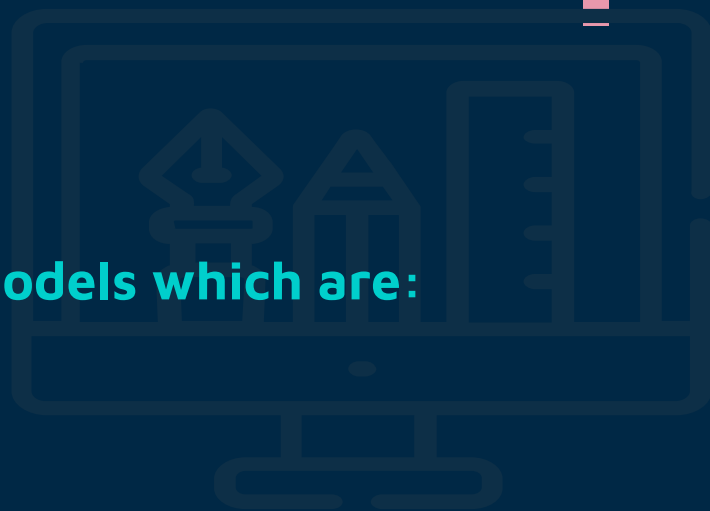
Train 10 different  
models

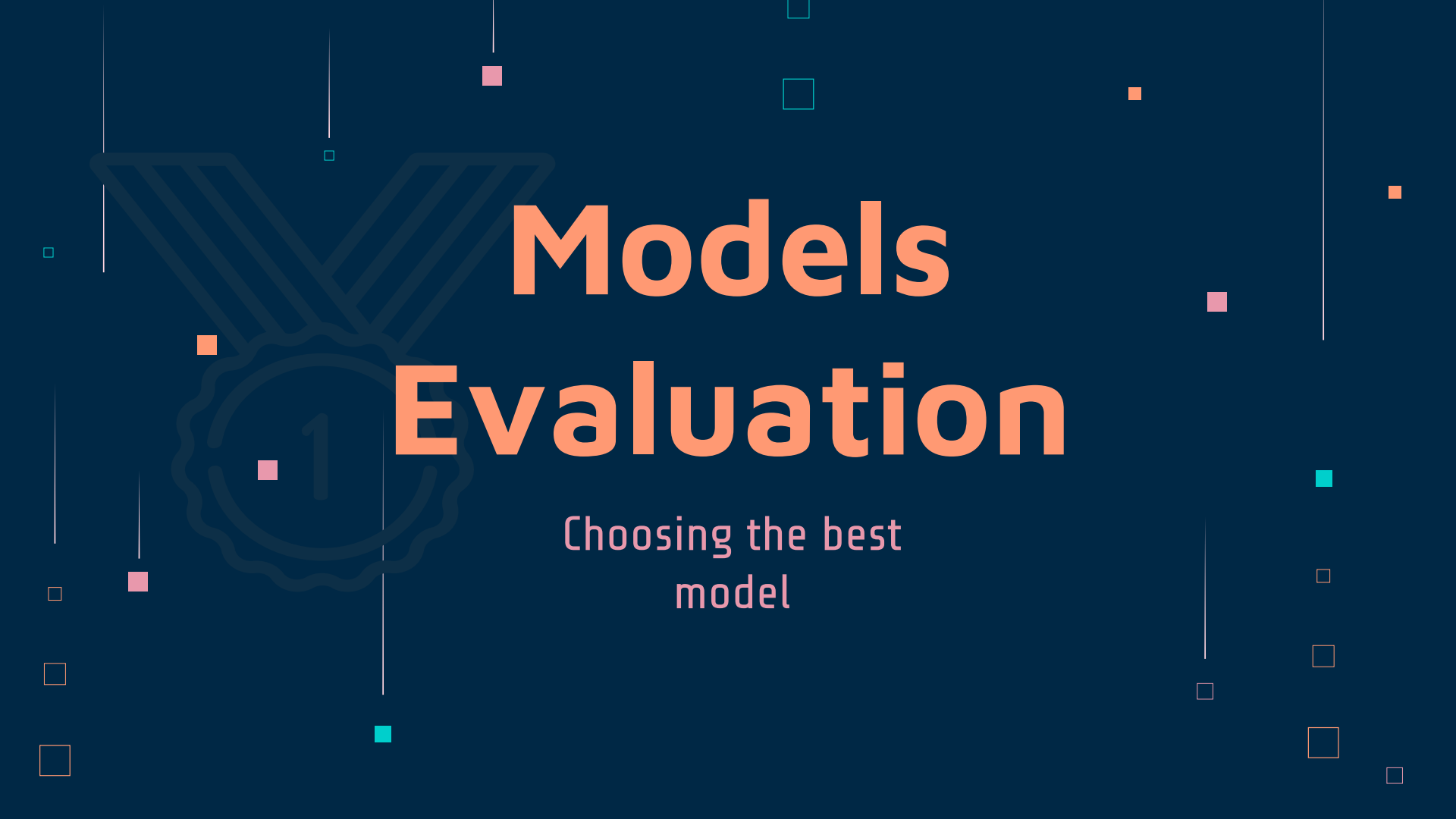
05

# Models training

**We start with training 10 models which are:**

- Random Forest regressor
- Extra Tree regressor
- Gradientboost regressor
- XGB regressor
- Linear regression
- Polynomial regression (1 to 5) degree





# Models Evaluation

Choosing the best  
model



06

## Compare between models Performance on the **training** And **cross Validation**

# Training

|   | Model                            | RMSE         | R2 Score |
|---|----------------------------------|--------------|----------|
| 0 | Random Forest                    | 7.280226e-01 | 0.994084 |
| 1 | Extra Trees Regressor            | 1.337159e-04 | 1.000000 |
| 2 | GradientBoost Regressor          | 2.002980e+00 | 0.955222 |
| 3 | XGB Regressor                    | 2.379262e-01 | 0.999368 |
| 4 | Linear Regression                | 4.202001e+00 | 0.802928 |
| 5 | Polynomial Regression (Degree 1) | 4.202001e+00 | 0.802928 |
| 6 | Polynomial Regression (Degree 2) | 2.707620e+00 | 0.918175 |
| 7 | Polynomial Regression (Degree 3) | 1.622808e+00 | 0.970607 |
| 8 | Polynomial Regression (Degree 4) | 4.919876e-13 | 1.000000 |
| 9 | Polynomial Regression (Degree 5) | 1.898513e-13 | 1.000000 |

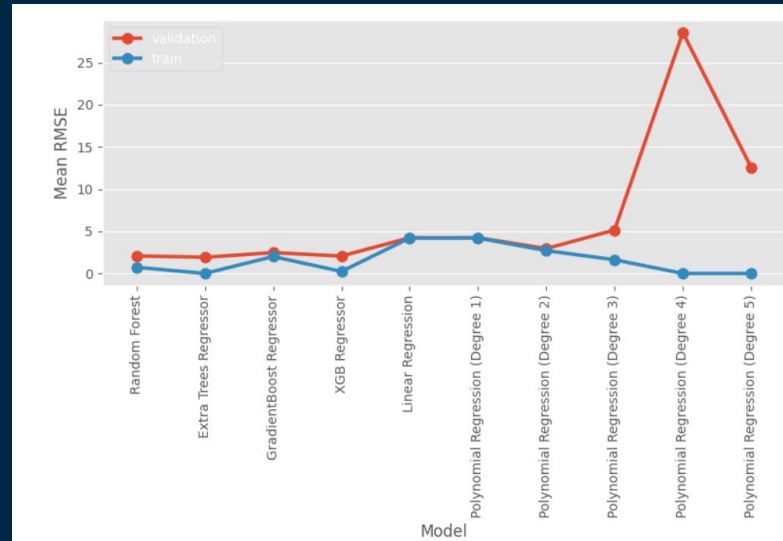
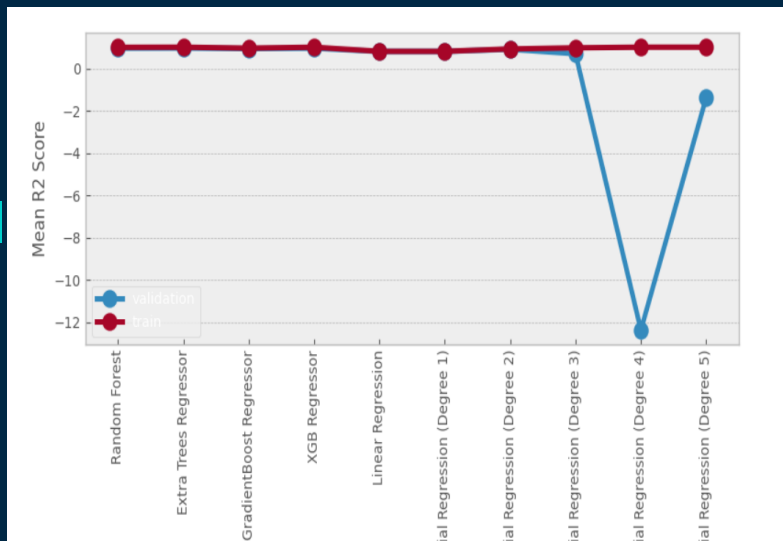
| Mean RMSE | Mean R2 Score |
|-----------|---------------|
| 2.080178  | 0.951072      |
| 1.918487  | 0.958538      |
| 2.466475  | 0.931106      |
| 2.059091  | 0.952201      |
| 4.233687  | 0.798767      |
| 4.233687  | 0.798767      |
| 2.947941  | 0.901861      |
| 5.146466  | 0.692076      |
| 28.562343 | -12.356042    |
| 12.580725 | -1.363345     |

# Cross Val

06

# Models Evaluation

Validation ■  
Training ■



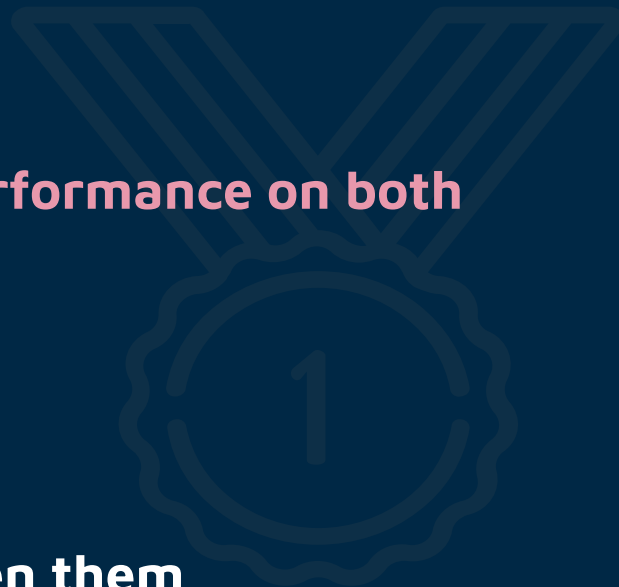
06

# Models Evaluation

**From the evaluation :**  
**some models show a high performance on both  
Training and Validation**

**Random Forest**  
**XGBoost**

**Which we can choose between them**



# 06

## Models Evaluation



**From the evaluation :**

**Other models perform bad on both the sets**

- **Linear Regression**
- **indication for Underfitting**

**Others perform very well on the training but  
performance Decreases on the validation**

**Higher order polynomials**

# Random Forest

Would be the Chosen  
model as it is the less  
shown overfitting

99%  
Training



95%  
Val



# Pipeline

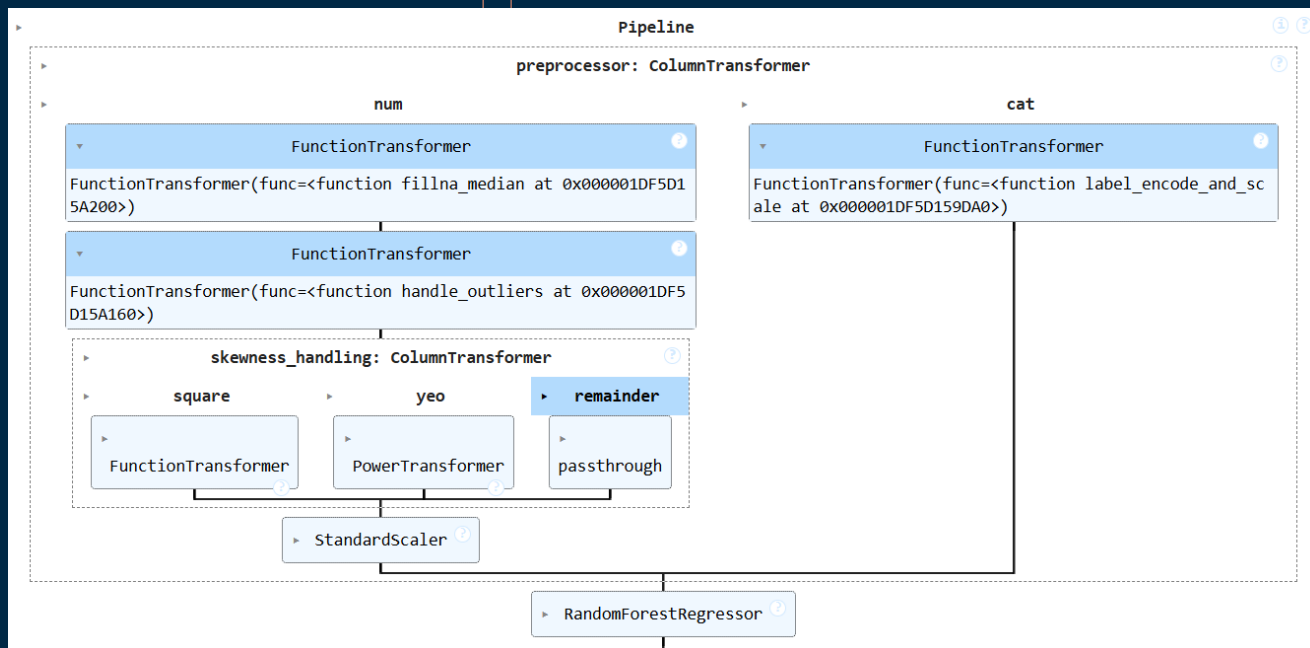
ALL IN ONE

# 07 Pipeline

Compress all the preprocessing , Cleaning , Transformation, Scaling, encoding, modeling and Predicting in one function

Takes row data as input and return prediction in one shot

# 07 Pipeline





# Hyperparameter Finetuning

Trying to finetune the  
performance

# 08 Hyperparameter finetuning

Using **grid search** we tried manually to set a combination of parameters

We fitted 5 folds for each 1920 Candidates which is overall number of Fits equal to 9400 fit

Unfortunately, the **RMSE** still the same

```
param_grid = {  
    'model__n_estimators': [100, 200, 300],  
    'model__max_depth': [None, 10, 20, 30],  
    'model__min_samples_split': [2, 5, 10],  
    'model__min_samples_leaf': [1, 2, 4],  
    'model__max_features': ['auto', 'sqrt'],  
    'model__bootstrap': [True, False]  
}
```

# Test Set

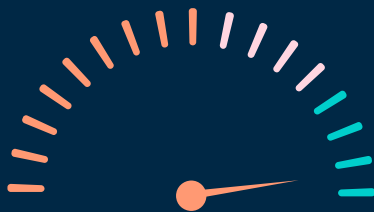
The Final Evaluation



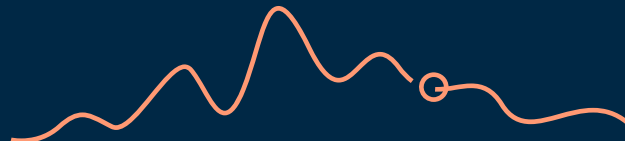
# 09

# Test Set

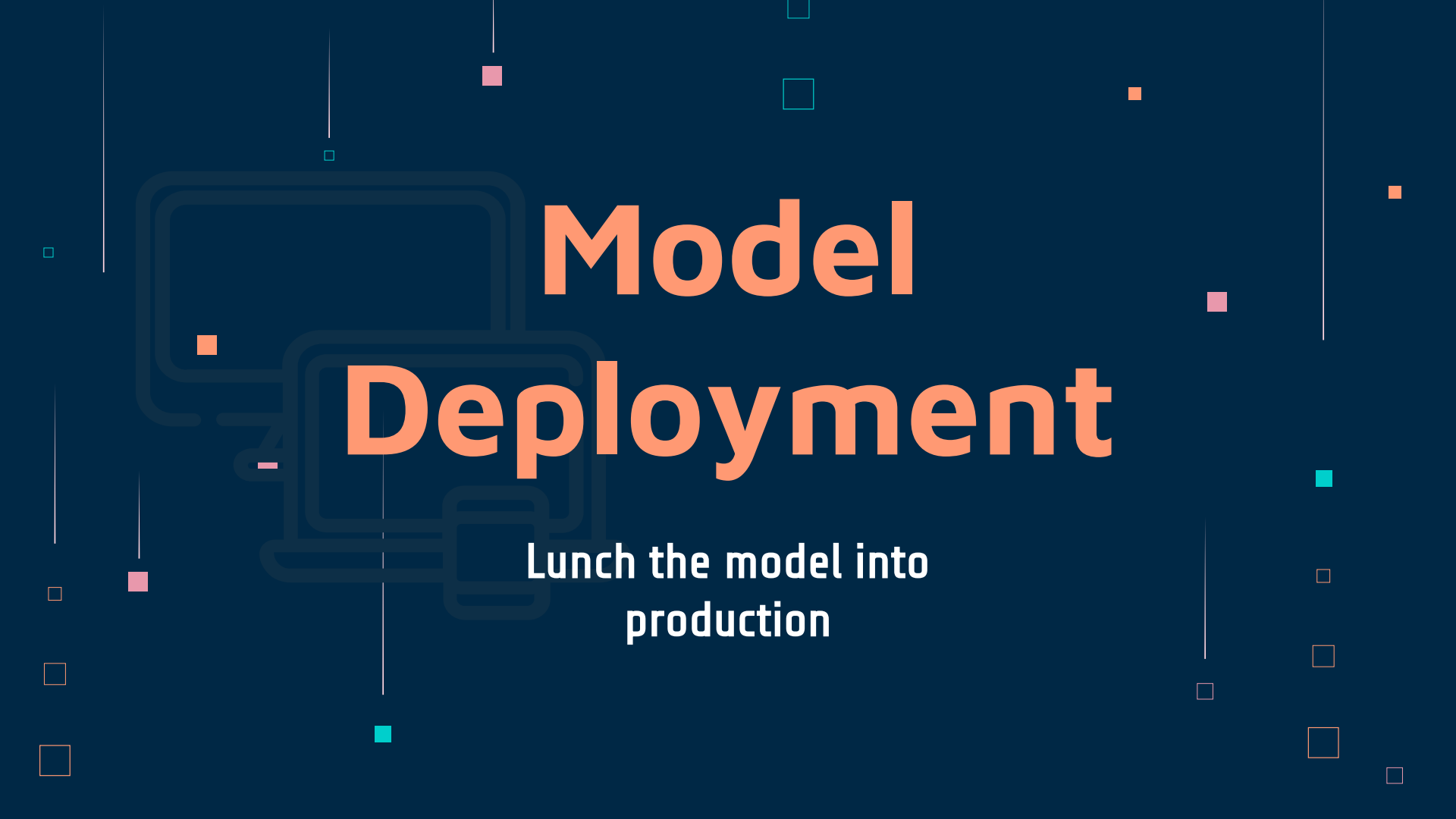
As we have a **pipeline**, we passed the test set to it  
And it gives us the **predictions directly**



R2 Score  
0.95



RMSE  
2.14



# Model Deployment

Lunch the model into  
production

# 08

## Saving & Deployment

We saved the model using **pickle module**  
And then deployed it using **Streamlit**

We created a **website** for the naïve user to enter the  
raw data and get the **prediction**

Country Life Expectancy Prediction

Survey Year

2006.00

-

+

Adult Mortality Rate

124.00

-

+

Infant Deaths Count

8.00

-

+

Alcohol Consumption Rate (liters per capita)

0.97

-

+

Hepatitis B Vaccination Coverage (%)

83.00

-

+

Measles Infection Count

517.00

-

+

Country Life Expectancy Prediction

Survey Year

2006.00

-

+

Adult Mortality Rate

123.00

-

+

Infant Deaths Count

8.00

-

+

Alcohol Consumption Rate (liters per capita)

0.97

-

+

Hepatitis B Vaccination Coverage (%)

83.00

-

+

Measles Infection Count

517.00

-

+

Gross Domestic Product (per capita, USD)

1762.25

-

+

Total Population

18914977.00

-

+

Thinness Rate (%)

6.38

-

+

Nation

Syrian Arab Republic

Country Category

Developing

Predict Life Expectancy

Predicted Life Expectancy: 69.91 years

# THANKS

Ahmed Mostafa Gamal El-din  
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Abdelrahaman Osama El-sayed  
Malak Mohamed Hassan Al-sharkawy  
Mallak Sadr El-din Hassan  
Aya Waleed Mohamed Shehata